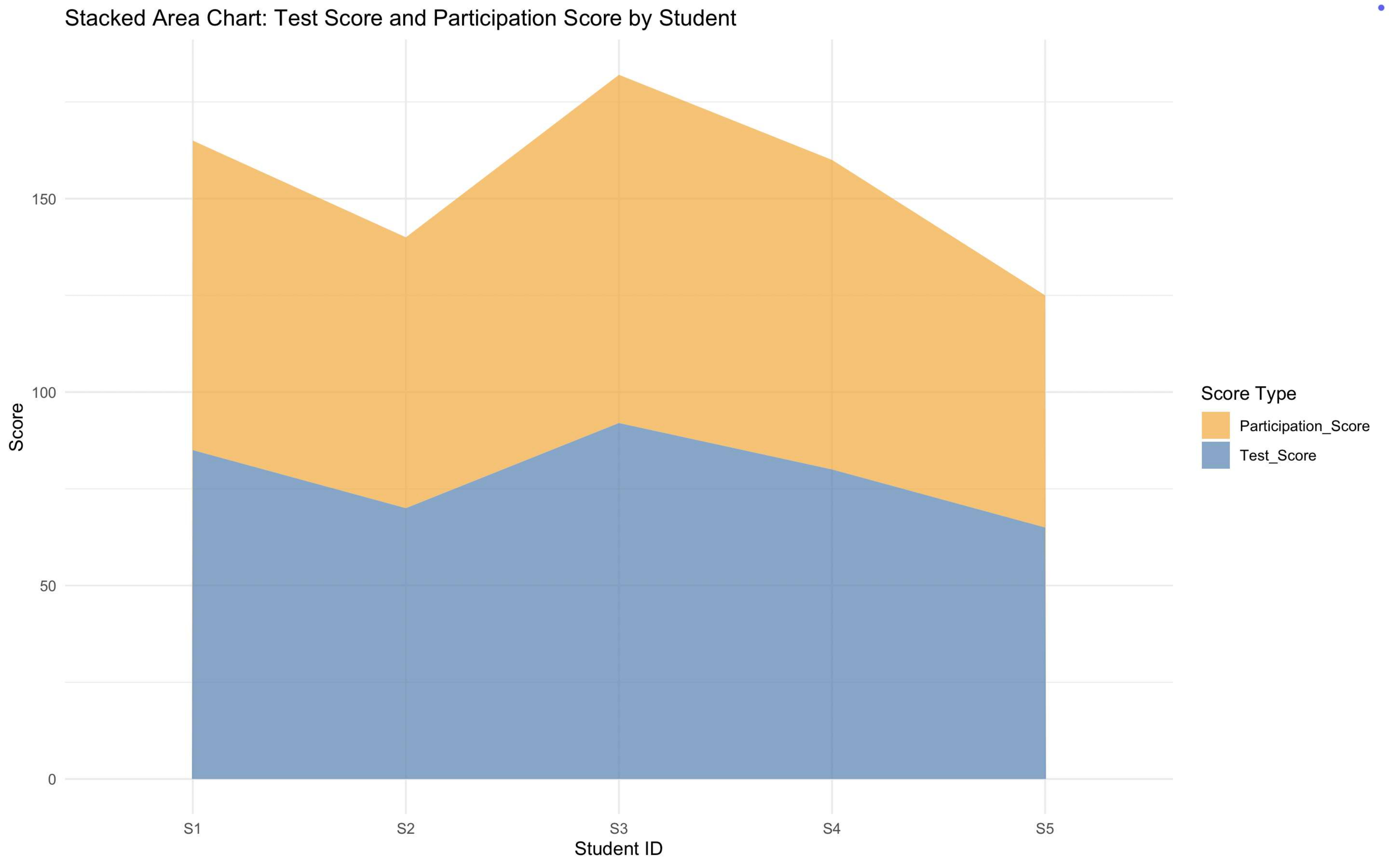
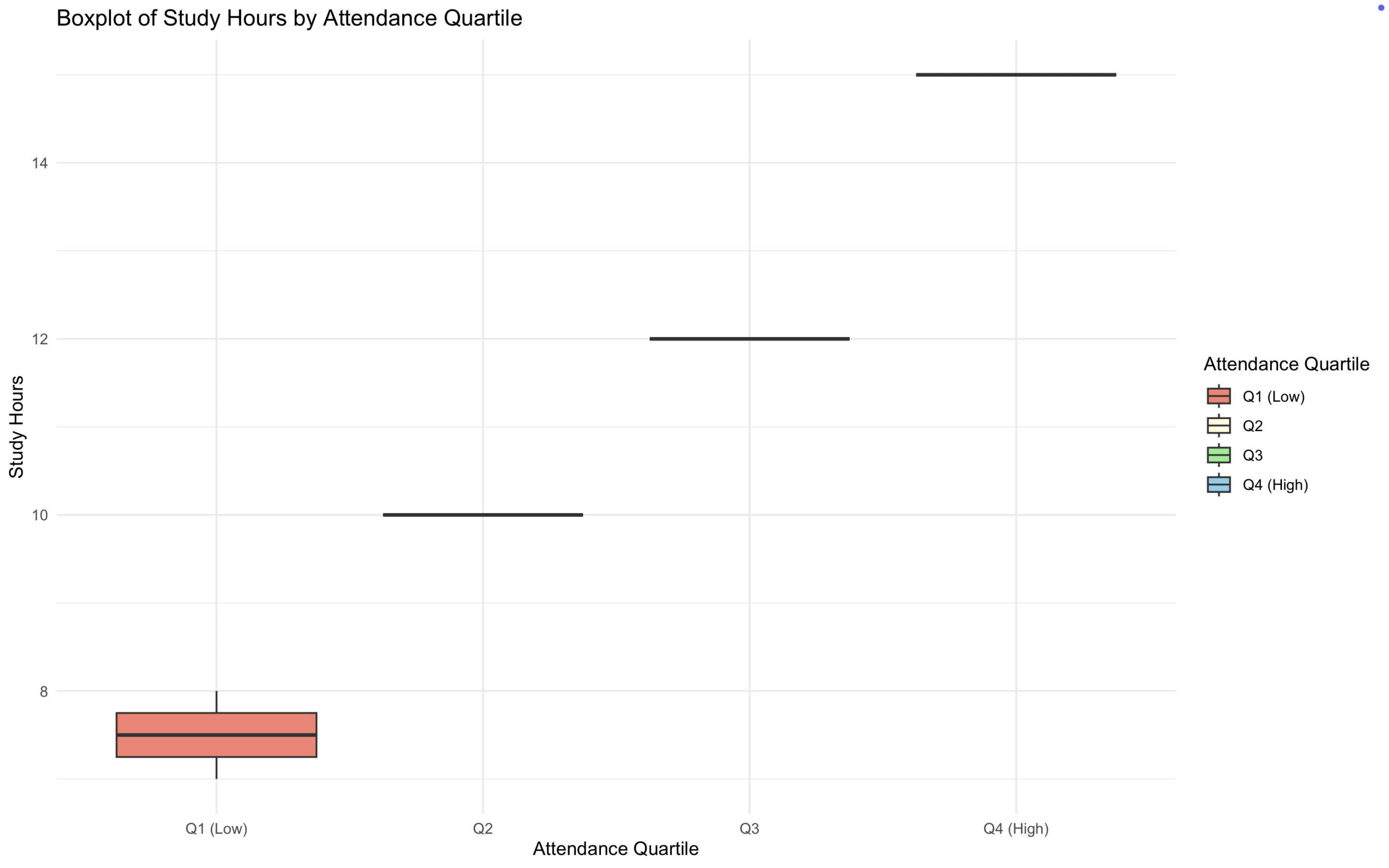
R Console

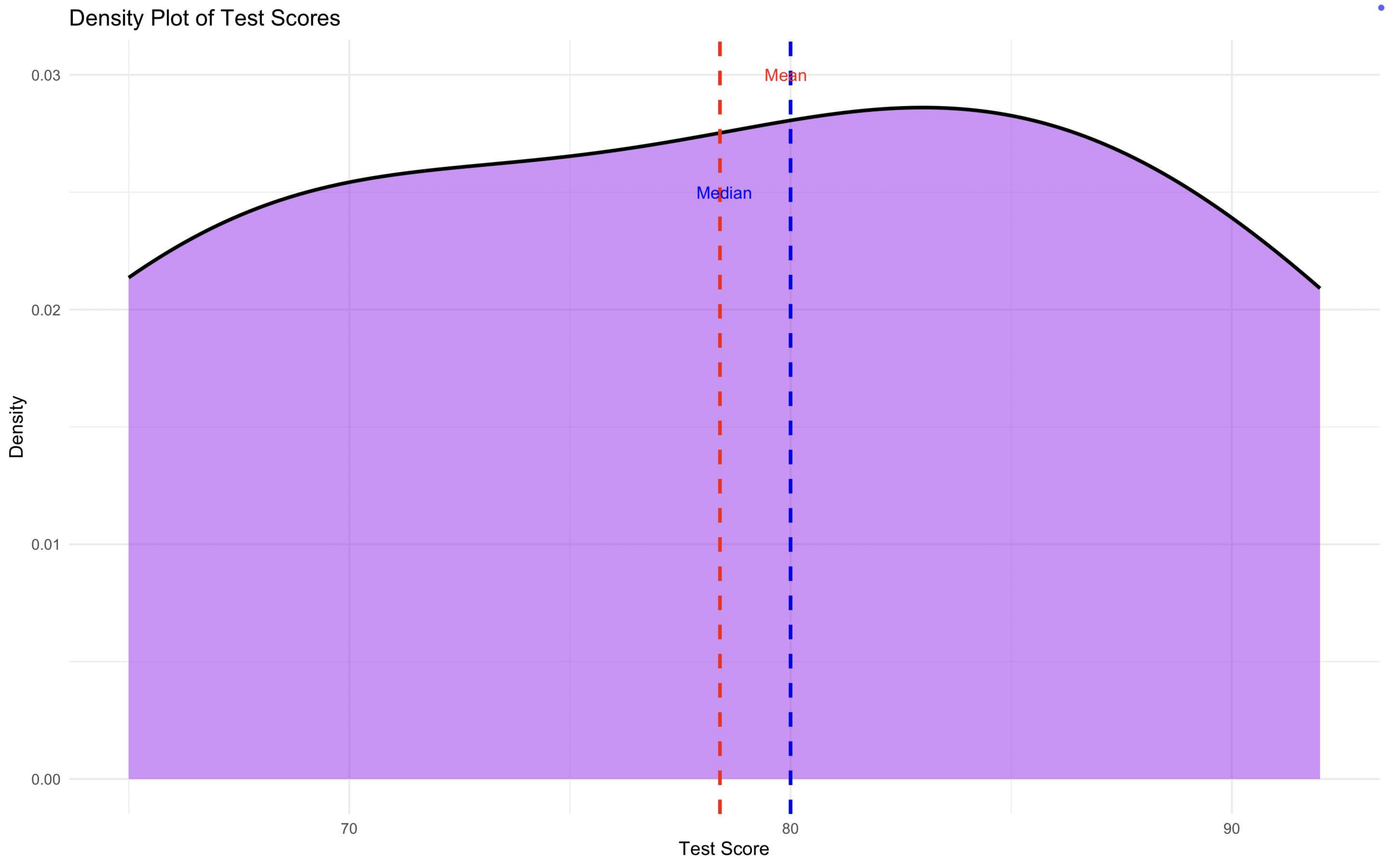
```
> # ----------------------------
> # Dataset Setup
> # ----------------------------
> student_data <- data.frame(
+   Student_ID = c("S1","S2","S3","S4","S5"),
+   Age = c(19, 21, 20, 22, 23),
+   Study_Hours = c(12, 8, 15, 10, 7),
+   Attendance = c(90, 70, 95, 85, 60),
+   Test_Score = c(85, 70, 92, 80, 65),
+   Participation_Score = c(8, 7, 9, 8, 6)
+ )
> 
> print(student_data)
  Student_ID Age Study_Hours Attendance Test_Score Participation_Score
1         S1  19          12         90         85                   8
2         S2  21           8         70         70                   7
3         S3  20          15         95         92                   9
4         S4  22          10         85         80                   8
5         S5  23           7         60         65                   6
> 
> library(ggplot2)
> 
> new_window <- function() {
+   if (.Platform$OS.type == "windows") {
+     windows()
+   } else {
+     quartz()
+   }
+ }
> 
> # ----------------------------
> # Q1: Stacked Area Chart - Test_Score and Participation_Score across Students
> # ----------------------------
> area_data <- data.frame(
+   Student_ID = rep(student_data$Student_ID, times = 2),
+   Score_Type = rep(c("Test_Score","Participation_Score"), each = nrow(student_data)),
+   Value = c(student_data$Test_Score, student_data$Participation_Score * 10)
+ )
> 
> new_window()
> print(
+   ggplot(area_data, aes(x = Student_ID, y = Value, fill = Score_Type, group = Score_Type)) +
+     geom_area(position = "stack", alpha = 0.7) +
+     scale_fill_manual(values = c("Test_Score" = "steelblue",
+                                  "Participation_Score" = "orange")) +
+     labs(title = "Stacked Area Chart: Test Score and Participation Score by Student",
+          x = "Student ID",
+          y = "Score",
+          fill = "Score Type") +
+     theme_minimal()
+ )
> 
> # ----------------------------
> # Q2: Boxplot - Study Hours by Attendance Quartile
```

R Console

```
+     labs(title = "Stacked Area Chart: Test Score and Participation Score by Student",
+          x = "Student ID",
+          y = "Score",
+          fill = "Score Type") +
+     theme_minimal()
+ )
>
> # ----------------------------
> # Q2: Boxplot - Study Hours by Attendance Quartile
> # ----------------------------
> student_data$Attendance_Quartile <- cut(
+   student_data$Attendance,
+   breaks = quantile(student_data$Attendance, probs = c(0, 0.25, 0.5, 0.75, 1)),
+   labels = c("Q1 (Low)","Q2","Q3","Q4 (High)"),
+   include.lowest = TRUE
+ )
>
> new_window()
> print(
+   ggplot(student_data, aes(x = Attendance_Quartile, y = Study_Hours,
+                            fill = Attendance_Quartile)) +
+     geom_boxplot(outlier.color = "red", outlier.size = 3) +
+     scale_fill_manual(values = c("Q1 (Low)" = "salmon",
+                                  "Q2" = "lightyellow",
+                                  "Q3" = "lightgreen",
+                                  "Q4 (High)" = "skyblue")) +
+     labs(title = "Boxplot of Study Hours by Attendance Quartile",
+          x = "Attendance Quartile",
+          y = "Study Hours",
+          fill = "Attendance Quartile") +
+     theme_minimal()
+ )
>
> # ----------------------------
> # Q3: Density Plot - Test Score Distribution
> # ----------------------------
> new_window()
> print(
+   ggplot(student_data, aes(x = Test_Score)) +
+     geom_density(fill = "purple", alpha = 0.5, color = "black", lwd = 1) +
+     geom_vline(aes(xintercept = mean(Test_Score)),
+                color = "red", linetype = "dashed", lwd = 1) +
+     geom_vline(aes(xintercept = median(Test_Score)),
+                color = "blue", linetype = "dashed", lwd = 1) +
+     annotate("text", x = mean(student_data$Test_Score) + 1.5,
+              y = 0.03, label = "Mean", color = "red", size = 3.5) +
+     annotate("text", x = median(student_data$Test_Score) - 1.5,
+              y = 0.025, label = "Median", color = "blue", size = 3.5) +
+     labs(title = "Density Plot of Test Scores",
+          x = "Test Score",
+          y = "Density") +
+     theme_minimal()
+ )
>
```


Stacked Area Chart: Test Score and Participation Score by Student
Score
150
100
50
0
S1
S2
S3
S4
S5
Student ID
Score Type
Participation_Score
Test_Score


Boxplot of Study Hours by Attendance Quartile
14
12
10
8
Study Hours
Q1 (Low)
Q2
Q3
Q4 (High)
Attendance Quartile
Attendance Quartile
Q1 (Low)
Q2
Q3
Q4 (High)


Density Plot of Test Scores
Mean
Median
0.03
0.02
0.01
0.00
Density
70
80
90
Test Score

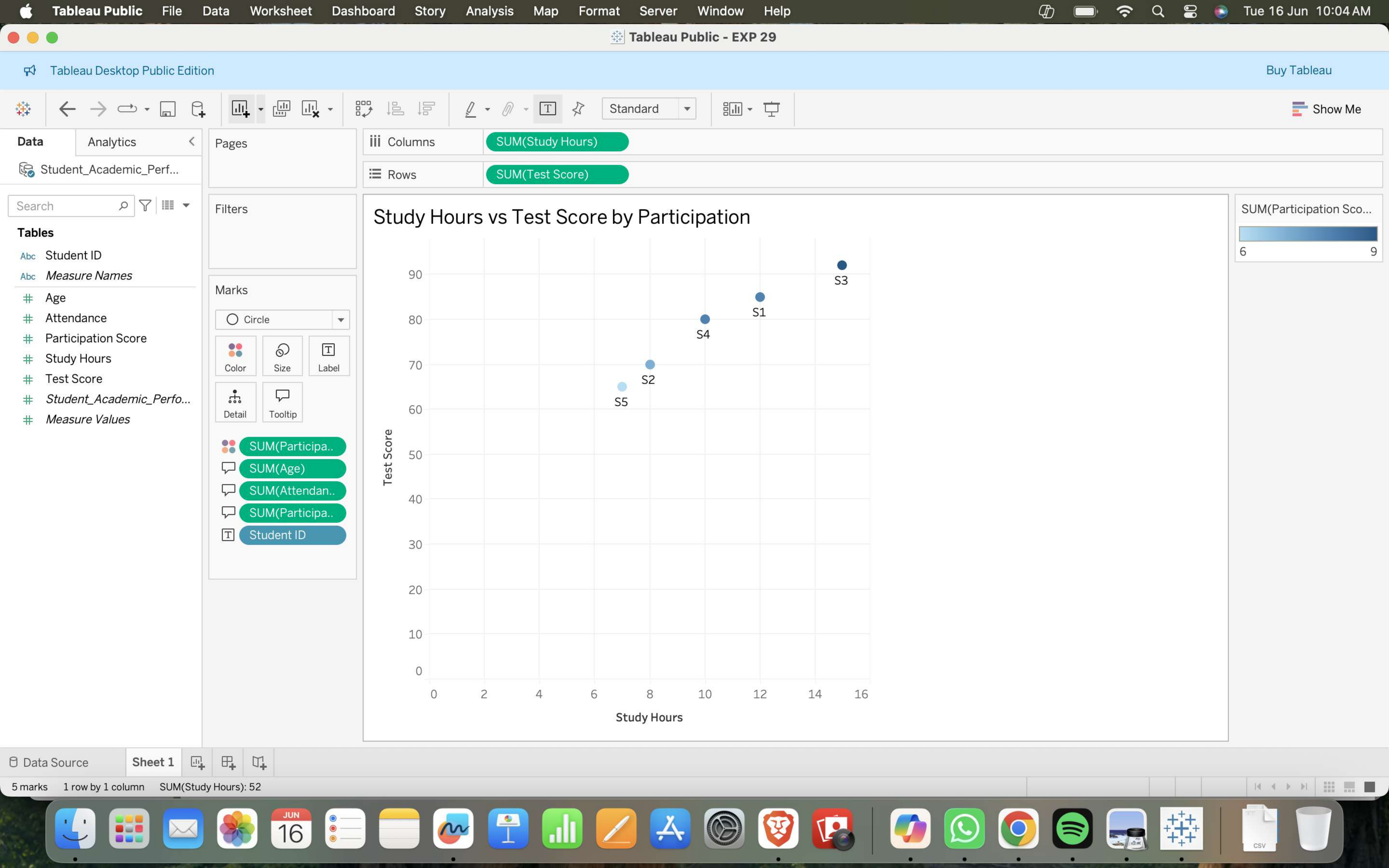


Tableau Public
File
Data
Worksheet
Dashboard
Story
Analysis
Map
Format
Server
Window
Help
Tue 16 Jun 10:04 AM
Tableau Public - EXP 29
Tableau Desktop Public Edition
Buy Tableau
Standard
Show Me
Data
Analytics
Student_Academic_Perf...
Search
Tables
Student ID
Measure Names
Age
Attendance
Participation Score
Study Hours
Test Score
Student_Academic_Perfo...
Measure Values
Pages
Columns
SUM(Study Hours)
Rows
SUM(Test Score)
Filters
Marks
Circle
Color
Size
Label
Detail
Tooltip
SUM(Participa..
SUM(Age)
SUM(Attendan..
SUM(Participa..
Student ID
Study Hours vs Test Score by Participation
SUM(Participation Sco...
6
9
S3
S1
S4
S2
S5
Test Score
90
80
70
60
50
40
30
20
10
0
0
2
4
6
8
10
12
14
16
Study Hours
Data Source
Sheet 1
5 marks
1 row by 1 column
SUM(Study Hours): 52
JUN
16
csv